

DATA ANALYST PORTFOLIO

Customer Churn Analysis Case Study

Transforming Customer Data into Actionable Insights Through Python Business Analysis, Visualizations, and Storytelling

Prepared by:

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Tools & Technologies:

Python (Pandas, Numpy) | Matplotlib/Seaborn| Jupyter Notebook | GitHub

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1. Summary

This project analyses customer churn patterns in a telecom-style dataset to identify key drivers of customer attrition and potential revenue loss. By exploring behavioral and contractual factors such as tenure, contract type, and pricing, the analysis uncovers actionable insights to improve retention strategies and reduce financial risk.

This analysis identifies high-risk churn segments and provides actionable recommendations to improve customer retention and reduce revenue loss.

2. Business / Analytical Question

This project was designed to answer four key business questions:

1. What factors drive customer churn?
2. Which customer segments are most at risk of leaving?
3. How does churn impact revenue?
4. What actions can the business take to reduce churn?

3. Dataset

The project uses the Churn dataset CSV file, which is uploaded to Jupyter notebook environment. The dataset contents include:

- Customer tenure
- Monthly charges
- Contract type (Month-to-month, One-year, Two-year)
- Churn status (Yes/No)
- Granularity: Customer-level data

4. Tools Used

- Python (Pandas, NumPy)
- Matplotlib/Seaborn
- Jupyter Notebook

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- Git & GitHub

Technical Methods Used:

- Data cleaning and transformation (Pandas)
- Aggregation and grouping
- Feature engineering (churn flag)
- Data visualization (Seaborn, Matplotlib)

5. Process (What I Did)

1. Data Cleaning
 - a. Handled missing values and inconsistent churn labels (True/False vs Yes/No)
 - b. Converted numerical fields and validated distributions
2. Exploratory Data Analysis (EDA)
 - a. Analyzed churn rates across different features
 - b. Grouped customers by tenure and contract type
 - c. Compared pricing behaviour between churned vs retained customers
3. Feature Exploration
 - a. Contract type vs churn
 - b. Tenure distribution vs churn
 - c. Monthly charges vs churn
4. Business Metrics Calculation
 - a. Estimated revenue at risk
 - b. Identified high-risk customer segments

6. Key Visualizations

1. Monthly Charges by Churn Status
 - Customers who churn tend to have higher monthly charges on average compared to retained customers
2. Churn Rate by Contract Type
 - Month-to-month customers have the highest churn rate (~40%), compared to ~10% for one-year and ~5% for two-year contracts.

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- Long-term contracts (1-year, 2-year) have much lower churn rates.
3. Tenure Distribution by Churn
 - Churn is heavily concentrated among customers with low tenure (0–10 months)
 - Long-tenure customers are far more likely to stay.
 4. Revenue at Risk
 - Month-to-month customers represent the largest revenue risk (~30k+)
 - Long-term contracts contribute minimal churn-related losses

7. Key Insights

1. Contract Type is the strongest churn driver
 - Customers without long-term commitment churn the most
2. Early customer lifecycle is critical
 - Most churn happens in the first few months
3. Pricing sensitivity matters
 - Higher-paying customers are more likely to leave
4. Revenue concentration risk
 - A disproportionate amount of revenue-risk comes from a specific segment (month-to-month users)

8. Business Implications and Recommendations (Why This Project Matters)

- Retention strategy focus:
Prioritize month-to-month customers with targeted offers
- Onboarding improvement:
Reduce early churn by improving customer experience in the first 90 days
- Pricing optimization:
Evaluate pricing fairness or perceived value
- Upselling opportunity:
Encourage migration to longer-term contracts

👉 Overall: Reducing churn in just one high-risk segment could significantly improve revenue stability.

9. Challenges and How to Solve Them

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Challenge 1: Inconsistent data labels

Solution: Standardized categorical values (Yes/No vs True/False)

Challenge 2: Duplicate or noisy features

Solution: Consolidated overlapping churn indicators

Challenge 3: Interpretation of visual overlaps

Solution: Used multiple visualizations (distribution + bar charts) for clarity

10. What I would Improve Next

To make the project even stronger in a production setting, I would:

- Build a predictive churn model (Logistic Regression/XGBoost)
- Perform feature importance analysis
- Add customer segmentation (clustering)
- Create an interactive dashboard (Power BI/Cognos)
- Include time-based churn trends

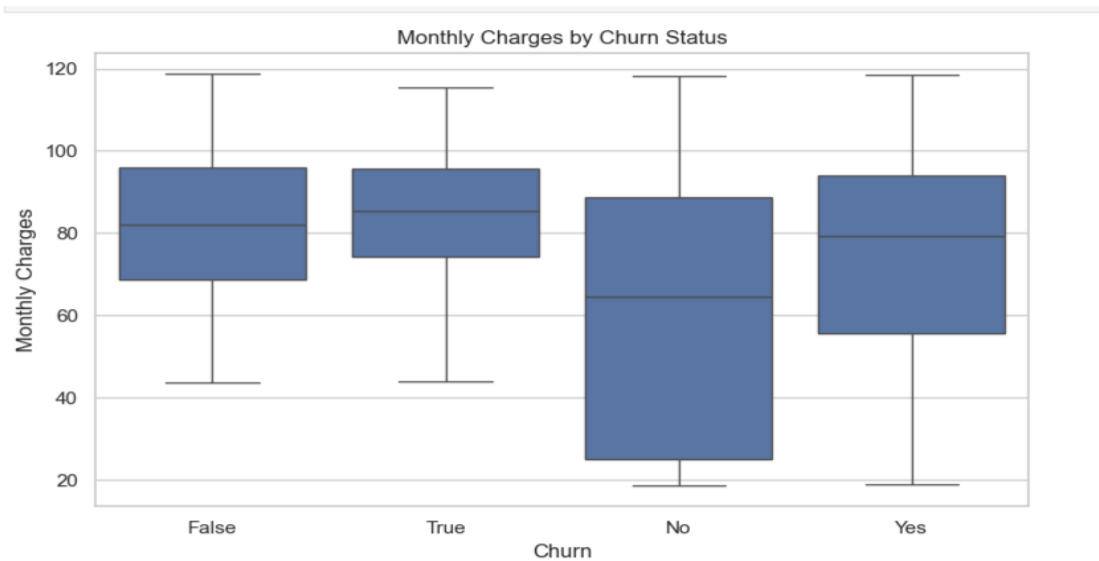
11. Files and Links

Include in repo:

- churn_data.csv
- Jupyter Notebook (analysis.ipynb)
- Charts (PNG/PDF)
- README.md

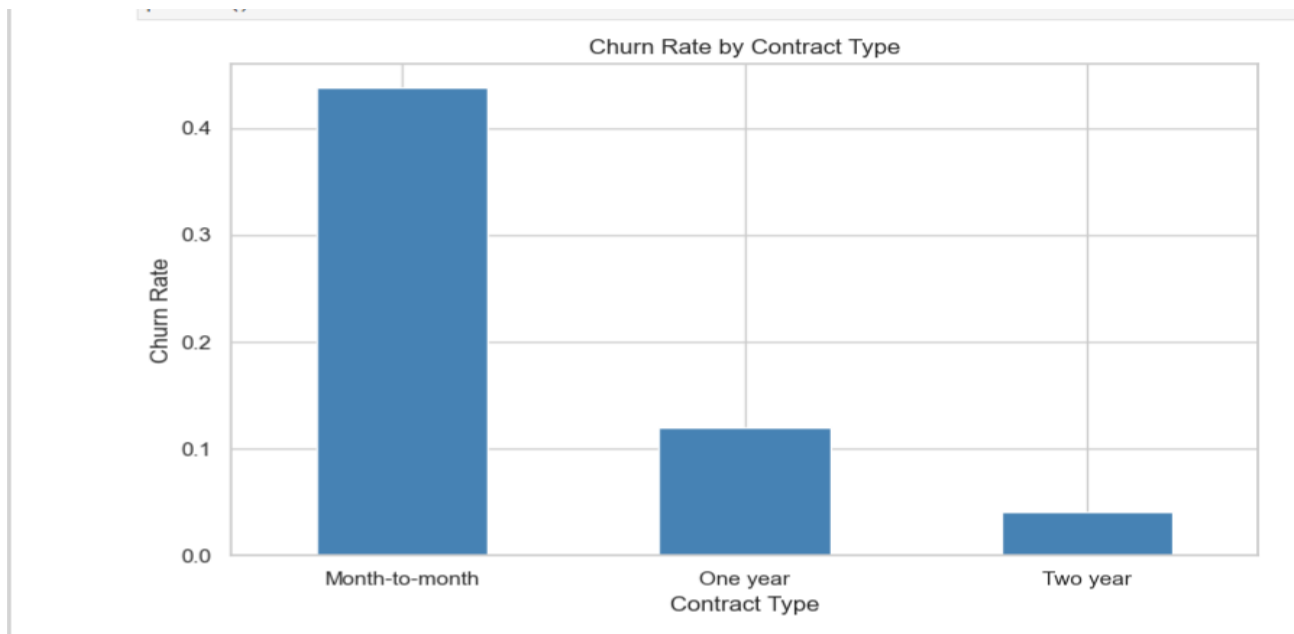
Chart 1: Monthly Charges by Churn Status

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Insight: Customers whose churn occupy more of the higher-charge range on a monthly charges scale, which runs up to **120**, supports the broader project findings that higher-paying customers are more likely to leave, suggesting potential pricing sensitivity.

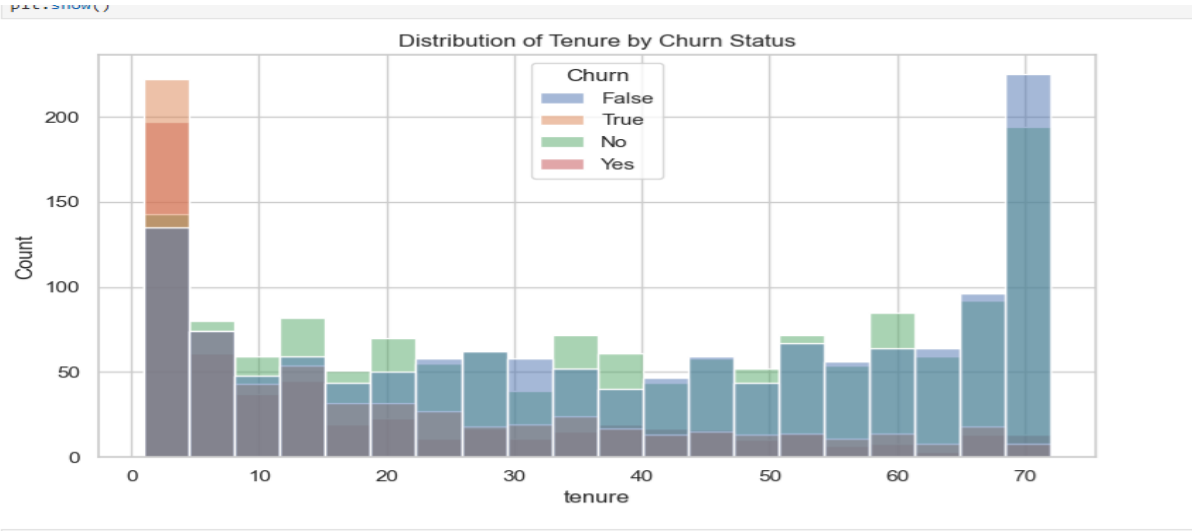
Chart 2: Churn Rate by Contract Type



Insight: Month-to-month customers have the highest churn rate at roughly **40%+**, while **one-year** and **two-year** contracts are substantially lower, showing that contract length is one of the strongest retention levers in the dataset.

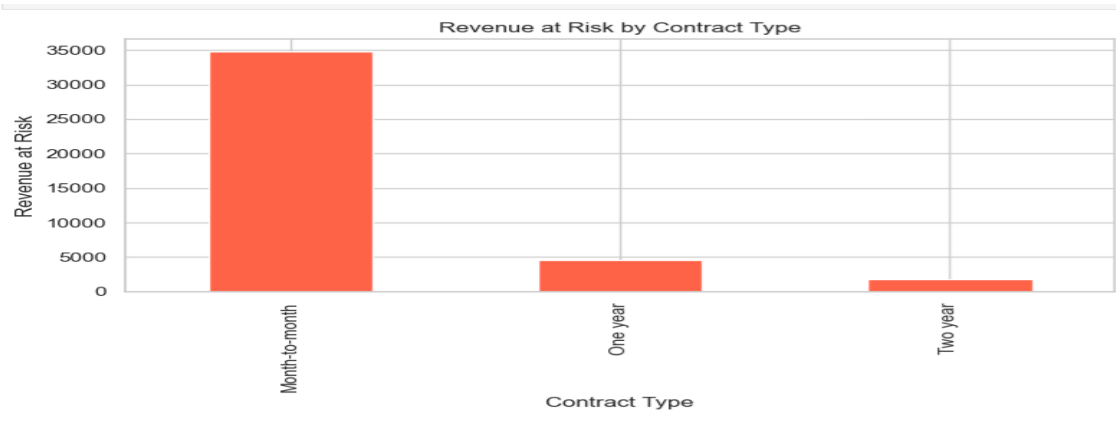
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Chart 3: Distribution of Tenure by Churn Status



Insight: Customer churn is heavily concentrated in the **0–10-month** tenure range, while long-tenure customers are far more likely to remain, indicating that the early customer lifecycle is the most critical period for retention.

Chart 4: Revenue at Risk by Contract Type



Insight: **Month-to-month** customers account for the largest revenue at risk at roughly **30k+**, while churn-related losses from longer-term contracts are comparatively small, showing that short-term customers represent the biggest financial exposure.